

Harish Manoharan

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Software engineer building production LLM systems (retrieval, model evaluation, AI agents) at Purgo AI, an AI startup, since 2025. M.S. Computer Science, UT Dallas (GPA 3.92/4.00); dual degree in Data Science and Biological Sciences, IIT Madras.

EXPERIENCE

Purgo AI

San Francisco Bay Area (Los Altos, CA)

AI startup building an LLM agent that writes and validates Databricks and dbt code

Software Engineer

Jun 2026 – Present

Software Engineer Intern

Jun 2025 – May 2026

- Reduced missed source tables by 70% (from 79 to 24 on a 103-ticket benchmark) in a LangGraph code-generation agent by building dbt lineage expansion (sqlglot SQL parsing, ref()/source() resolution), chosen through a four-variant ablation.
- Raised table-retrieval precision from 0.914 to 0.947 with zero recall regressions (95 tickets) by adding a pre-filter that checks candidate tables against warehouse DDL before the LLM reranker.
- Cut LLM cost of the retrieval and drafting stages 7.6x at equal recall (0.944 vs 0.931 over 5 runs of 102 tickets) by moving them to a smaller reasoning model and tuning reasoning effort per stage.
- Improved table-search recall from 0.54 to 0.68 and cut latency from 4.0 s to 0.18 s (188 real queries) by building semantic search on Pinecone for Polaris/Iceberg catalogs with incremental indexing.
- Delivered an LLM evaluation and validation product that qualifies Azure OpenAI and Databricks-served models for GxP use: 19 evaluation tests for hallucination, accuracy/F1, bias, determinism, p95 latency and traceability.
- Built most of the IQ/OQ validation engine for Databricks on Azure and AWS: 59 versioned tests in 17 suites, sandboxed execution, auditor evidence reports and an LLM pipeline turning requirement docs into tests.
- Improved generated PySpark/SQL reliability by extending the agent's LangGraph orchestration with a self-review loop (AST lint rules for Databricks runtime failures, recovery, best-candidate selection).

Digitus Business Solutions

Bengaluru, India

Software Engineer Intern

May 2024 – Jul 2024

- Raised SQL execution accuracy from 52% to 76% on a 500-query evaluation set by building a text-to-SQL agent with schema discovery and retries driven by execution errors.
- Built a multimodal RAG assistant over internal slide decks and videos (layout-aware chunking, enforced citations).

PwC India

Bengaluru, India

Technology Consulting Intern, Data Analytics

May 2023 – Jul 2023

- Cut forecast error (MAPE) 15% against the baseline across 50+ SKUs with an XGBoost + SARIMA ensemble; built an aspect-based sentiment pipeline over 30k+ customer reviews.

EDUCATION

The University of Texas at Dallas M.S. Computer Science

May 2026

GPA 3.92/4.00 · Dean's Graduate Scholar

Indian Institute of Technology Madras Dual Degree: M.Tech Data Science, B.S. Biological Sciences

May 2024

INSPIRE Scholar · Young Research Fellow · Thesis with The Jackson Laboratory on unsupervised behavior discovery in mice

SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Machine learning and LLMs: retrieval-augmented generation (RAG), LLM evaluation, AI agents (LangGraph), embeddings and vector search (Pinecone, pgvector), LLM observability (Langfuse), PyTorch

Data: Databricks, Apache Spark (PySpark), dbt, PostgreSQL, Apache Iceberg

Backend and cloud: FastAPI, NestJS, Next.js, REST APIs, GCP, AWS, Docker, CI/CD (GitHub Actions)

Work authorization: F-1 OPT through June 2027; STEM OPT-eligible to mid-2029.